LALITH VARA PRASAD KARAMPUDI

Software Engineer — Python, Backend Services, MLOps & Applied ML

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PROFESSIONAL SUMMARY

Recent Computer Science graduate (B.Tech, Data Science specialization) who builds, tests, deploys and documents backend and ML software end to end: validated REST APIs (FastAPI), Docker deployments, automated tests and GitHub Actions CI, experiment tracking and model monitoring. Designs for failure (leases, retries, guardrails, drift checks) and reports results honestly, including negative ones. Every project below has a runnable demo or walkthrough in my portfolio. Published researcher (ICACTEA-2025); Oracle Foundations Associate certified.

TECHNICAL SKILLS

Languages: Python (primary), SQL, JavaScript, Java and C++ (coursework), Git/GitHub

Backend & Deployment: FastAPI, REST API design, Redis, queue reliability (leases, retries, idempotency, dead-lettering), Docker, docker-compose, GitHub Actions CI, Render

ML / AI: Scikit-learn, XGBoost, PyTorch, TensorFlow, MLflow, Optuna, SHAP, RAG, GraphRAG, LLM agents and guardrails, Ollama, ChromaDB, NetworkX

Engineering Practice: pytest (unit, integration, offline mocks), failure/chaos testing, debugging and root-cause analysis, documentation

Certifications: Oracle Foundations Associate

PROJECTS (live demos for each at lalith0087.github.io)

Task Pipeline — Crash-Tolerant Job Queue | Python, FastAPI, Redis, Docker Compose, GitHub Actions | github.com/lalith0087/Task-pipeline

- Two services (API + scalable workers) on Redis with atomic Lua dequeue, heartbeat leases, fencing tokens, exponential backoff, dead-letter queue and idempotency keys.
- 21 tests against real Redis plus a manual kill -9 test in which another worker recovered the crashed worker's job; CI runs the tests and a Compose smoke test.

Guarded RAG Agent — Tool-Calling Agent with Guardrails | Python, FastAPI, Ollama, ChromaDB, pydantic, Docker | github.com/lalith0087/rag-agent-guardrails

- Framework-free agent with schema-validated tools, injection screening, grounding checks and an optional LLM judge; served via FastAPI with a full step trace and chat UI.
- 31 offline tests and CI. On an 11-question eval with a 1B local model, the judge-on setting shipped 0 wrong answers by abstaining when unsure.

Fraud Detection MLOps | Python, Scikit-learn, MLflow, FastAPI, Docker, GitHub Actions | github.com/lalith0087/fraud-detection-mlops

- Class-weighted gradient boosting on 284,807 real transactions (0.17% fraud): PR-AUC 0.735 / ROC-AUC 0.962 on a stratified held-out set, using PR-AUC because of the class imbalance.
- Cost-based threshold tuning cut expected cost per 1,000 transactions by ~14% versus the F1-optimal threshold (34.15 to 29.23) at recall 0.88.
- MLflow tracking with a gated model registry, PSI drift monitoring, and a FastAPI service in Docker with CI (6 tests).

Telco Customer Churn — Deployed ML Pipeline | Python, Scikit-learn, XGBoost, MLflow, Optuna, SHAP, FastAPI, Streamlit, Render | github.com/lalith0087/telco-churn-mlops

- Tuned XGBoost model (ROC-AUC 0.84) across 75+ MLflow/Optuna experiments, served by FastAPI with a SHAP explanation endpoint and LLM-written plain-English summaries.
- Deployed API and a Streamlit UI on Render, so anyone can enter a customer profile and get a prediction plus a SHAP chart (live link on the repo).

Second Brain GraphRAG — Local Knowledge-Graph Assistant | Python, Ollama, ChromaDB, NetworkX, FastAPI | github.com/lalith0087/second-brain-graphrag

- Private GraphRAG system: LLM extraction of entities and relations into a NetworkX graph beside a ChromaDB index; answers with cited facts via 2-hop expansion.
- Benchmarked against plain RAG on 62 chunks (198 entities, 165 relations): no accuracy gain at 1B-model scale (6/16 each). Ruled out prompt truncation as the cause and published the negative result.

Local RAG Assistant — Chat with a Codebase | Python, ChromaDB, Ollama, FastAPI | github.com/lalith0087/local-rag-assistant

- Fully local retrieval-augmented assistant with AST-aware chunking of Python source, multi-repo indexing, cited answers and a model-agnostic LLM layer.
- Traced a hallucination failure to its root cause and documented the mitigation; 21 tests and CI.

Image Captioning using CNN + LSTM | Python, TensorFlow, PyTorch | github.com/lalith0087/IMAGE-CAPTIONING-using-CNN-and-LSTM

- Led a team through design, training and evaluation; co-authored the paper published at ICACTEA-2025, Aditya University (April 2025).
- Re-implemented in PyTorch with a reproducible BLEU eval (BLEU-1 0.498, BLEU-4 0.128 on held-out Flickr8k, 20 epochs), a split-fingerprint check against leakage, 7 tests and CI.

Fast-AgingGAN — Face Aging | Python, PyTorch Lightning | github.com/lalith0087/Face-aging-

- Lightweight GAN trained on CACD and UTKFace with a Lightning training loop; repo ships a pretrained generator and an inference script that writes before/after grids.

More projects: RSVP Movies (SQL and Python analysis of IMDb data with an executive summary); Home Credit Risk EDA (pandas and seaborn analysis of loan applicants); JavaScript Mini Projects (52 small browser apps, games and API tools); Text Encryptor (key-based cipher in JavaScript). All at github.com/lalith0087.

PUBLICATION

"Image Captioning using CNN and LSTM," ICACTEA-2025, Aditya University — April 2025.

EDUCATION

B.E. / B.Tech — Computer Science (Data Science) | Vignan Institute of Technology and Science | 2020–2025 | CGPA 6.9/10

Coursework: Machine Learning, Statistical Methods, Data Structures, Python, Data Science Libraries, OOP.

Intermediate (MPC), Narayana Junior College, 2018–2020, 70% | SSC, Lords Group of Schools, 2008–2018, 9.2 CGPA